Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] **[14EE81]**

EIGHTH SEMESTER B.Tech EXAMINATION, JUNE–2019
POWER SYSTEM OPERATION AND CONTROL

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Answer any **FIVE** full Questions. Choosing by Atleast two questions from each part.
2. Missing data may be suitably assumed.

PART - A

1. a. Discuss briefly about the control features and functions of control centre. **07**

b. Discuss about the SCADA functions in a power system. **07**

2. a. With a neat block diagram explain the brushless Excitation systems. **07**

b. Explain about Load compensator and Field shorting circuits **07**

3. a. Explain how mathematical model of speed governor system is developed for automatic generation Control (ACG). **09**

b. A power system has a load of 1250MW at 50 Hz, D=1.5 If 50 MW is tripped, find the steady state frequency deviation when; **05**

i. There is no speed control

ii. The system has reserve of 200 MW spread over 500 MW of generation capacity with 5% regulation on this capacity. All other generators are operating with valves wide open. Due to dead band only 80% of governors respond to load change.

4. a. Explain briefly the components/equipment's of power system that can generate and/or absorb reactive power. **04**

b. Briefly explain the terms Voltage stability, Voltage collapse and factors causing voltage instability. **04**

c. In the radial transmission system shown in Fig.4(c), determine the real and reactive powers to be delivered by the generator G. Neglect the voltage drops in the lines and transformers. All P.U values are referred to the voltage bases shown and 100MVA. Determine also the power factor at which the generator must operate. **06**

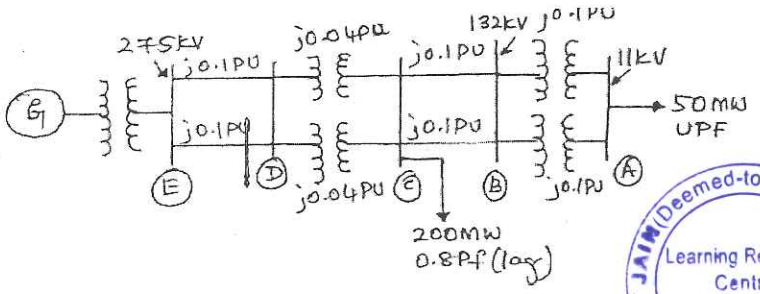

Fig-4c.

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PART - B

5.	a.	Discuss the thermal constraints and the role of spinning reserve in unit commitment Problem.	05
	b.	Explain the priority list method of unit commitment problem with an example.	09
6.	a.	Explain the contingency analysis for detection of network problems.	06
	b.	Explain the optimal dispatch, post contingency, Secure dispatch and secure post contingency states of power system with an example.	08
7.	a.	With a neat flow chart, explain the contingency analysis using sensitivity factors.	07
	b.	With a neat flow chart, explain the 1P1Q contingency selection procedure.	07
8.	a.	Explain the three modes of failure of a system.	05
	b.	Briefly explain the two state generator models. With the usual notations derive the expression for availability and unavailability in terms of failure and repair rate.	05
	c.	Explain the calculation procedure for Generating System Reliability Evaluation	04